

Enhancing Bone Regeneration and Osseointegration using rhPTH(1-34) and Dimeric $R^{25}C$ PTH(1-34) in an Osteoporotic Beagle Model

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Abstract

Introduction

Despite continuous parathyroid hormone (PTH) exposure potentially accelerating bone resorption, intermittent PTH administration has shown anabolic effects on bone microarchitecture. This study investigates the therapeutic impacts of two PTH analogs, rhPTH(1-34) and dimeric $R^{25}C$ PTH(1-34) on bone regeneration and osseointegration in a postmenopausal osteoporosis animal model.

Methods

Twelve female beagles, osteoporotic post-ovariectomy, underwent implant surgeries. Animals were divided into three groups: control, daily rhPTH(1-34) injection, and daily dimeric $R^{25}C$ PTH(1-34) injection. After 10 weeks, bone regeneration and implant osseointegration were evaluated using micro-CT, histological/histomorphometric analyses, and serum biochemical analysis.

Results

While the rhPTH(1-34) group demonstrated improved microarchitectural characteristics, such as BMD, BV, trabecular metrics, and osseointegration degree, the dimeric $R^{25}C$ PTH(1-34) group showed similarly enhanced anabolic effects around the titanium implants, albeit these were less pronounced than the rhPTH(1-34), yet significantly better than the control group. Histologic and TRAP analyses revealed that both PTH analogs significantly promoted bone regeneration and remodeling, especially in artificially created bone defects.

Conclusion

This study demonstrated the therapeutic effects of rhPTH(1-34) and dimeric ^{R25C}PTH(1-34) on bone regeneration and titanium osseointegration in a beagle model with osteoporosis.

eLife assessment

This work by Shin et al. demonstrated that a different form of PTH (R25C PTH) generated a comparable anabolic signal to rhPTH 1-34 using a large animal model. This **valuable** finding may have therapeutic potential in promoting bone formation or the healing process, and the methods seem **solid**.

Introduction

Since excessive levels of circulating parathyroid hormone (PTH) increase osteoclastic activity and accelerate bone resorption, it might seem paradoxical that PTH can also be used as a treatment modality for diseases with bone loss, such as osteoporosis (Jilka, 2007). Intermittent administration of PTH unlikely continuous exposure showed anabolic effects, indicating different responses relating to bone microarchitecture depending on the dose and frequency (Silva and Bilezikian, 2015). However, the underlying mechanisms remain largely unknown, although it was shown that PTH binds through PTH type 1 receptor (PTH1R) and that G-protein coupled receptors (GPCR) are associated with the protein kinase A (PKA)-dependent pathway, thereby demonstrating primary anabolic action on bone (Cheloha et al., 2015; Jilka, 2007). The anabolic effect of intermittent PTH administration is mediated by the downregulation of the Wnt/beta-catenin signaling pathway, which upregulates the transcriptional expression of growth factors, such as IGF1, FGF2, and Runx2, which are essential for the proliferation and differentiation of osteoblasts, leading to an increased number of osteoblasts and survival (Krishnan et al., 2006; Lee et al., 2009).

A PTH analog, teriparatide (rhPTH(1-34)), demonstrated its promising anabolic effects in a fracture prevention trial (FPT) (Neer et al., 2001), leading to its approval by the United States Food and Drugs Administration, making it the first anabolic agent for postmenopausal osteoporotic women. Additionally, its unique anabolic features, which are contrasted with antiresorptive, lead to an increased application of rhPTH(1-34), whereby it was used in both metabolic and pathologic bone diseases alongside various other conditions where bone formation occurred (Uusi-Rasi et al., 2005). Conversely, recent concerns regarding the development of osteonecrosis of the jaw (ONJ) have appeared in association with the use of antiresorptive, meaning rhPTH(1-34) has gained attention for its potential to reduce the risk of ONJ and its therapeutic effects in treating ONJ (Jung et al., 2017; Kakehashi et al., 2015). Although in vivo studies on fracture healing, bone augmentation, and titanium osseointegration effects of rhPTH(1-34) have been attempted, they have generally been limited to pilot studies in rodents (Gomes-Ferreira et al., 2020; Jung et al., 2021; Yu and Su, 2020).

Interestingly, only nine mutations have been discovered since the PTH amino acid and nucleotide sequences were confirmed (Lee et al., 2020; Schipani et al., 1999). Among them, the PTH R25C mutation was discovered in three siblings with familial idiopathic hypoparathyroidism, which consisted of a homozygous arginine to cysteine mutation at residue 25 (R25C) in the mature PTH(1-84) polypeptide and exhibited distinct characteristics from the others (Lee and Lee, 2022). The other mutations located in the prepro-leader region of the hormone resulted in defective synthesis

and secretion; however, the PTH R25C mutation is located within the mature bioactive domain of PTH and does not affect synthesis or secretion ([Lee and Lee, 2022](#)). Although the capacity of R25CPTH(1-34) to bind to the PTH1R and stimulate cAMP production was slightly lower in the human osteoblast-derived SaOS-2 cell line ([Lee and Lee, 2022](#)), yet it showed comparable anabolic activity in a mouse model ([Bae et al., 2016](#)). Furthermore, the dimeric formation of the R25CPTH(1-84) peptide, presumably through disulfide bonding of the cysteine residues, has implied that dimeric R25CPTH(1-34) might partake in unique biological actions, which potentiate its clinical applications ([Park et al., 2021](#)).

In addition to the limited effectiveness toward certain types of fractures, such as non-vertebral fractures, and the disadvantage of its limited duration of use, the administration of rhPTH(1-34) for osteoporosis has raised concerns about its potential to induce cortical porosity, despite showing favorable results in the treatment of trabecular microarchitecture ([Lindsay et al., 2016](#)). This has led to concerns regarding the widespread use of PTH. However, recent notable studies have indicated that cortical porosity is not induced when PTH is administered weekly ([Mosekilde et al., 1995](#)), which is in contrast to daily administrations. ([Yamamoto et al., 2016](#); [Zebaze et al., 2017](#)) These studies suggest that the frequency and dosage of PTH administration can significantly affect the bone response ([Hock et al., 1992](#)), while different forms of PTH might also induce different biological responses, which means deeper investigations into the therapeutic effects of PTH are required ([Bellido et al., 2005](#)). Therefore, in this study, the authors used a large animal model that mimics postmenopausal osteoporosis to investigate the therapeutic effects of two PTH analogs, rhPTH(1-34) and dimeric R25CPTH(1-34), on bone regeneration and osseointegration.

Results

Microarchitectural and histological analysis of titanium osseointegration

Various characteristics of the right mandible were evaluated in this analysis, including bone mineral density (BMD), bone volume (BV), trabecular number (Tb.N), trabecular thickness (Tb.Th), and trabecular separation (Tb.S). ([Figure 2](#)) The group administered PTH(1-34) presented consistently higher values of BMD, BV, Tb.N, and Tb.Th., and lower value of Tb.S compared to the control and (Cys25)PTH(1-84) groups, indicating that PTH(1-34) administration enhanced titanium osseointegration. Moreover, this was consistent for all three titanium implants, irrespective of artificial bone defects (second implant) or bone grafting (third implant). Interestingly, the (Cys25)PTH(1-84) group showed similar trends for the anabolic effects related to the titanium implants. Morphometric analysis indicated that (Cys25)PTH(1-84) administration enhanced the BMD, BV, and other trabecular indices, resulting in a higher degree of osseointegration than in the control group ($P < 0.05$), although lower than in the PTH(1-34) group ($P > 0.05$).

Histological analysis further clarified the micro-CT results. The implants in the control group showed physiological bone osseointegration around the titanium implant; however, insufficient bone–implant contact and exposure of implant threads were observed. ([Figure 3](#)) This was especially evident on the buccal side of the implant, which was more vulnerable due to the bundle bone structure. Alternatively, the titanium implants in the PTH(1-34) group were in full contact with the green-stained mineralized bone. The (Cys25)PTH(1-84) group showed a better pattern of osseointegration compared to the control group, although the bone–implant contact was lower than in the PTH(1-34) group. Notably, both the PTH(1-34) and (Cys25)PTH(1-84) groups presented evidence of bone regeneration for the second implant, whereby a bone defect was created prior to placing the implant. However, both were insufficient compared to the control group and did not illustrate any bone filling. The measured bone–implant contact ratio was $18.32 \pm 16.19\%$ for the control group, $48.13 \pm 29.81\%$ for the group, and $39.53 \pm 26.17\%$ ($P < 0.05$).

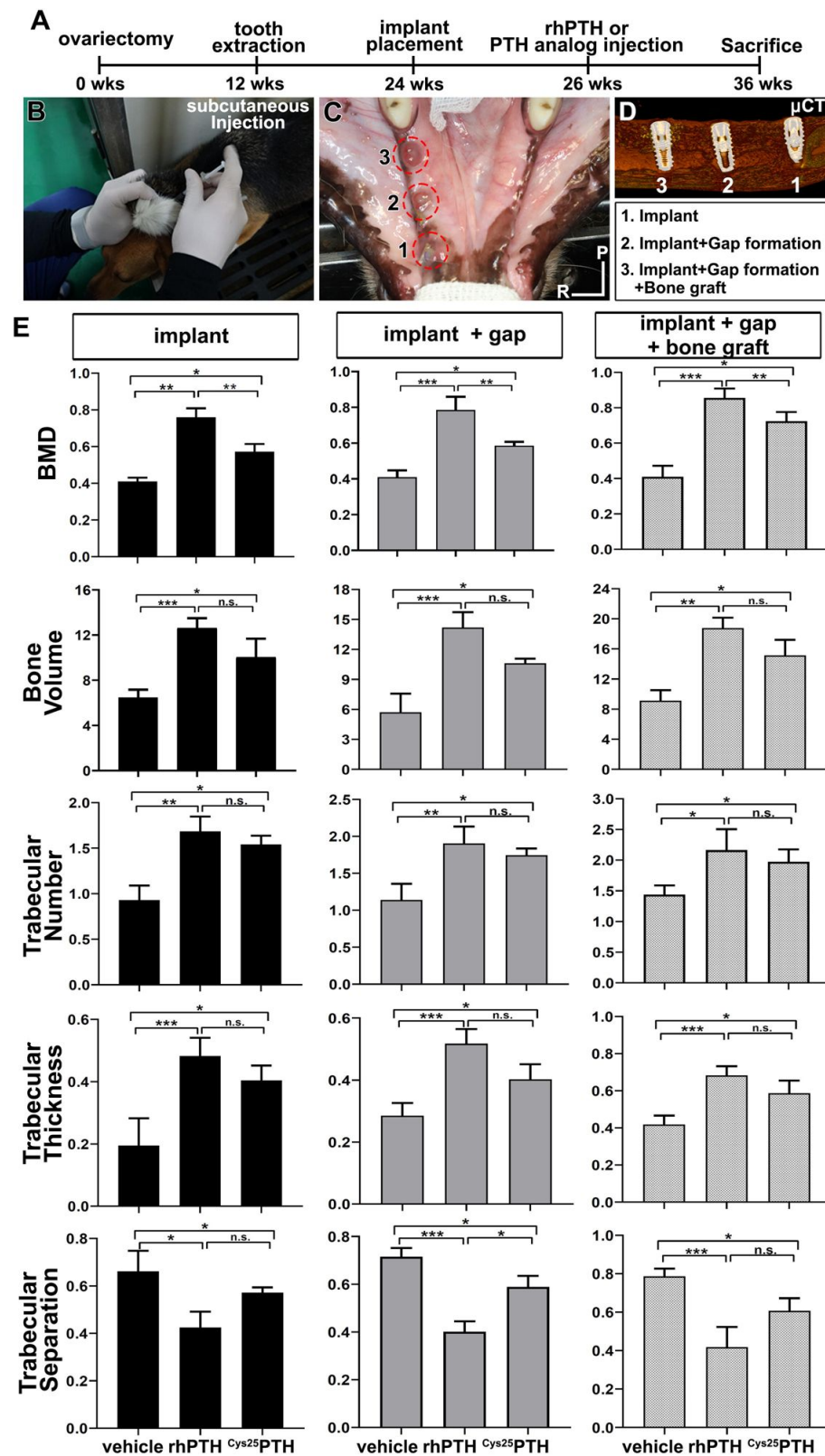


Figure 1.

A) Experimental timeline, **B)** subcutaneous injection of PTH analogs, **C)** intraoral photograph, **D)** micro-CT scan of implant inserted in the right mandible.

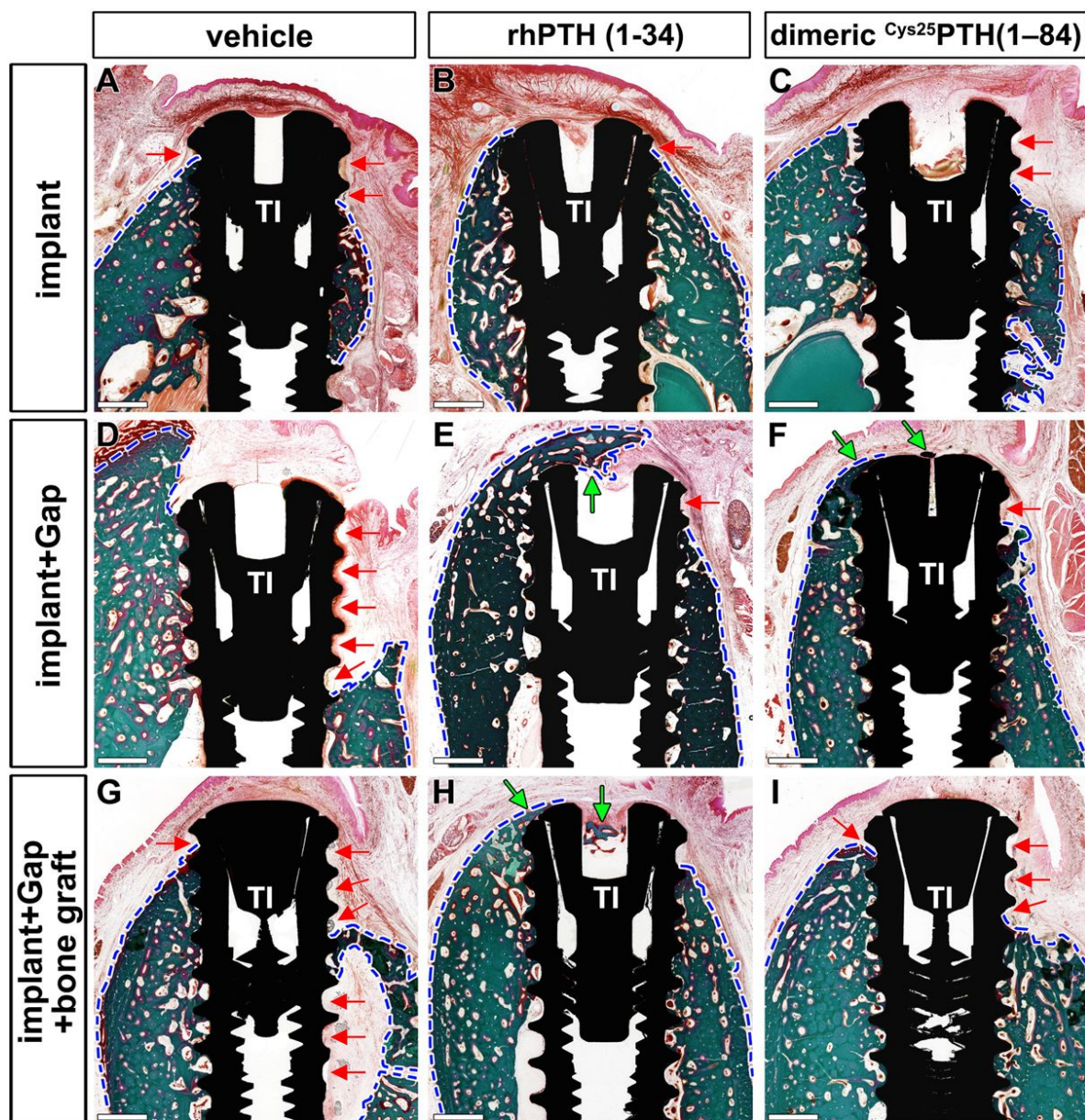


Figure 2.

Microarchitectural CT analysis.

(**A**) Bone mineral density (BMD), (**B**) bone volume (BV; mm³), (**C**) trabecular number (Tb.N; 1/mm), (**D**) trabecular thickness (Tb.Th; μm), (**E**) trabecular separation (Tb.S; μm).

* indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$

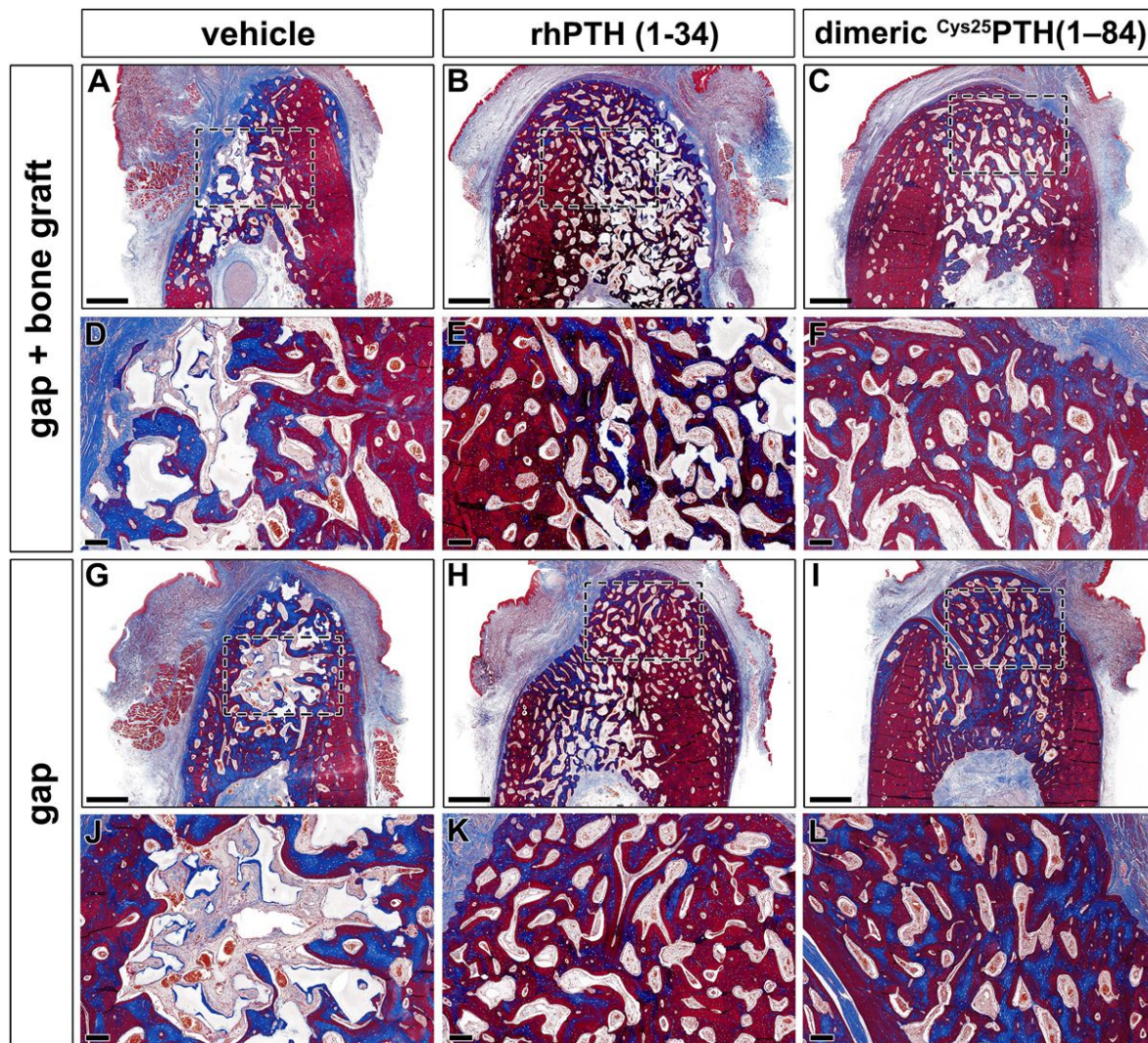


Figure 3.

(1) Histological analysis using Goldner's trichrome staining for the regenerated bone around the titanium implant (right mandible), (2) histological analysis using Masson trichrome staining for the bone remodeling pattern (left mandible).

Histological and TRAP analyses of bone regeneration

Artificial bilateral bone defects were created in the left mandible and the effects of the two PTH analogs were evaluated on the bone regeneration, with one left unfilled and the other filled with a bone graft. **Figure 4** demonstrates the effects of PTH(1-34) and (Cys25)PTH(1-84) on bone regeneration compared to the control group. Following the formation of the bone defects, the PTH(1-34) and (Cys25)PTH(1-84) groups achieved sufficient morphological bone regeneration over a period of 10 weeks, while the control group exhibited morphological incompleteness over the same period. The PTH(1-34) group exhibited a mature trabecular architecture, while the (Cys25)PTH(1-84) group showed a similar morphology, although some immature bone formation remained stained blue in the Masson trichrome staining analysis. While there was no clear difference in the bone defect between the sites with and without bone grafting, the site where bone grafting occurred exhibited a more mature bone morphology, indicating that a xenograft-maintained space for new bone formation with osteoconductive effects.

The capability of PTH(1-34) and (Cys25)PTH(1-84) in bone remodeling was evaluated by TRAP immunohistochemical staining. (**Figure 5**) Both the PTH(1-34) and (Cys25)PTH(1-84) groups showed a significantly higher number of TRAP+ cells at both bone defects, with and without a xenograft, compared to the control group. ($P < 0.05$) In addition, the number of TRAP+ cells in the (Cys25)PTH(1-84) group was significantly higher than in the vehicle, yet lower than in the PTH(1-34) group.

Serum biochemical analysis

The levels of calcium, phosphorus, CTX, and P1NP were analyzed over time using RM-ANOVA. There were no significant differences between the groups for calcium and phosphorus at time points T0 and T1. However, after the PTH analog was administered at T2, the levels were highest in the PTH(1-34) group, followed by the (Cys25)PTH(1-84) group, and then, lowest in the control group, which was statistically significant. ($P < 0.05$) The differences between the groups over time for CTX and P1NP were not statistically significant.

Discussion

This study investigated the therapeutic effects of rhPTH(1-34) and dimeric R^{25C} PTH(1-34) on bone regeneration and osseointegration in a large animal model with postmenopausal osteoporosis. rhPTH(1-34) and dimeric R^{25C} PTH(1-34) have shown significant clinical efficacy, and although there have been a few studies investigating their effects on bone regeneration in rodents ([Garcia et al., 2013](#)), the authors in this study aimed to investigate the effects using a large animal model that more accurately mimics osteoporotic humans ([Cortet, 2011](#)). In the evaluation of titanium osseointegration, the PTH(1-34) group consistently exhibited enhanced bone mineral density (BMD), bone volume (BV), and other key parameters, thereby indicating superior titanium osseointegration compared to the control and dimeric R^{25C} PTH(1-34) groups. Histological analyses confirmed these results, emphasizing the stronger bone-implant contact observed in the rhPTH(1-34) group. Furthermore, both PTH analogs significantly promoted bone regeneration in artificially created defects, with the rhPTH(1-34) group displaying a more mature trabecular architecture, as evidenced by a notable increase in the TRAP+ cell count during the bone remodeling assessments.

Furthermore, by demonstrating that dimeric R^{25C} PTH(1-34) exhibits a distinct pharmacological profile different from rhPTH(1-34) but still provides a clear anabolic effect in the localized jaw region, the authors have shown that it may possess different potential therapeutic indications from rhPTH(1-34).

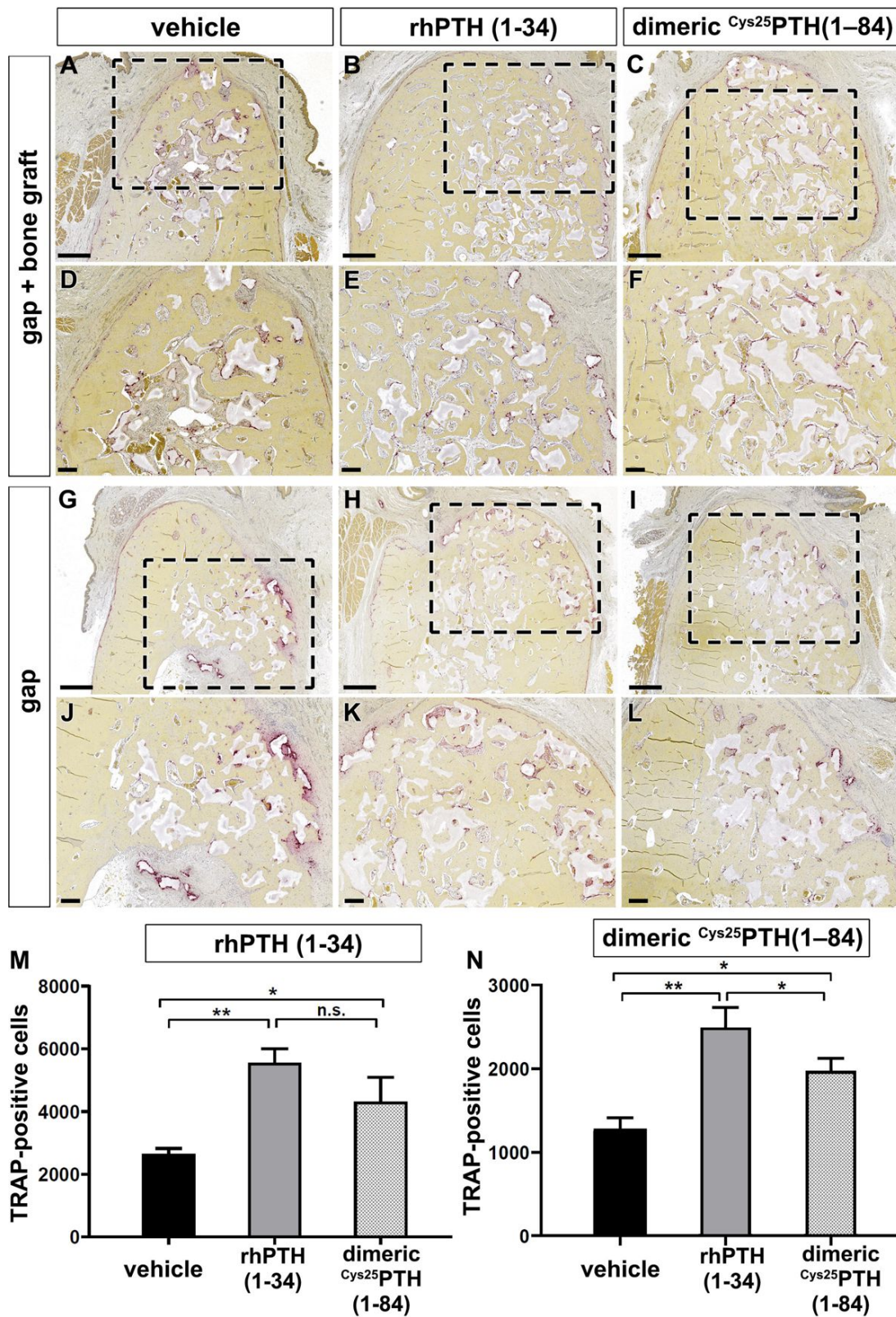


Figure 4.

Immunohistochemical analysis using TRAP staining for bone remodeling activity. (A-L) TRAP-positive cells (M-N) indicates $p < 0.05$, ** indicates $p < 0.01$, and *** indicates $p < 0.001$.

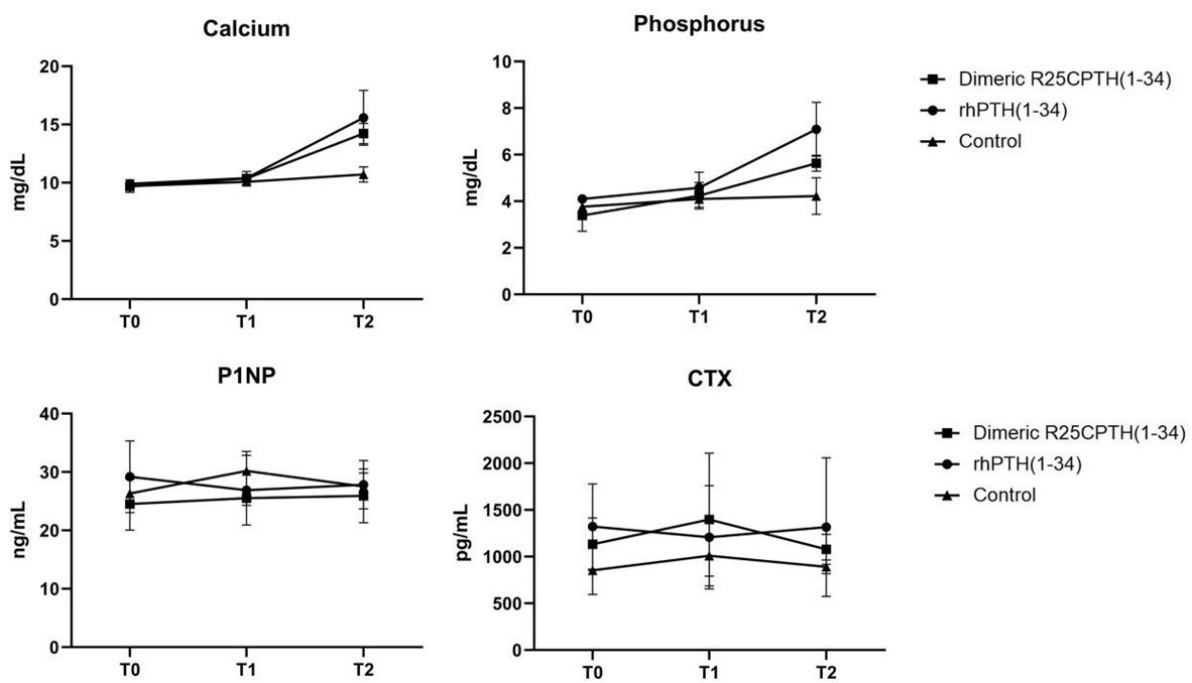


Figure 5.

Serum biochemical analysis.

In addition to the effect of systemic bone mineral gain, the unique anabolic feature of PTH has received clinical attention as an emerging strategy due to the increased risk of ONJ following the use of antiresorptive and the rising need for implantation and bone augmentation in the field of orthopedics and maxillofacial surgery (Ruggiero *et al.*, 2022). Moreover, site-specific differential effects of teriparatide have not been clarified, although it represents an issue often associated with selective concentrations of teriparatide that cause anabolic effects on the central skeleton and possible bone mineral decrease on the peripheral skeleton, including the skull (McClung *et al.*, 2005). Moreover, it can be inferred that facial and jaw bones, which have the same developmental origin as the skull through membranous ossification, will show the same bone response as the skull (Setiawati and Rahardjo, 2019). Previous studies have demonstrated that the central skeleton, including the lumbar and thoracic spine and pelvis regions, showed an increase in areal BMD, while the arms, legs, and skull showed a decrease in bone minerals, suggesting an effect of PTH on the redistribution of bone minerals from the peripheral to the central skeleton (Paggiosi *et al.*, 2018).

However, the results of this study demonstrated that rhPTH(1-34) and dimeric R^{25C} PTH(1-34) significantly improved the osseointegration of bone and titanium, as well as jawbone regeneration. The authors have attributed this phenomenon to the unique anatomical characteristics observed in the jawbone. The jawbone in the human body undergoes the most rapid bone remodeling and has excellent blood flow (Huja *et al.*, 2006). since is continuously exposed to mechanical stress due to mastication and swallowing, thereby suggesting that the net anabolic effect of PTH in the jawbone, which is not part of the central skeleton, is achieved through mechanical loading. A recent study by Robinson *et al.* demonstrated that PTH and mechanical loading additively stimulate anabolic modeling and synergistically stimulate remodeling in trabecular bone, findings that further support this notion (Robinson *et al.*, 2021). However, further investigation is needed to fully understand this relationship.

The anabolic effects of PTH have been demonstrated through several large randomized controlled trials (Neer *et al.*, 2001; Tsai *et al.*, 2013), although the drawbacks arising from the unique characteristics of PTH have hindered its broad clinical application. In addition to the disadvantages of continuous self-injection, high cost, and a two-year treatment limit, the greatest concern with prolonged use of teriparatide might be accelerated bone remodeling, subsequent hypercalcemic condition, and increased bone resorption (Burr *et al.*, 2001; Fox, Miller, Newman, *et al.*, 2007; Fox, Miller, Recker, *et al.*, 2007; Neer *et al.*, 2001; Sato *et al.*, 2004; Tsai *et al.*, 2013). However, recent studies have shown that the frequency and dosing of PTH administration can lead to different bone responses. (Yamamoto *et al.*, 2016; Yamane *et al.*, 2017). An *in vivo* study by Yamamoto *et al.* reported that lower doses of rhPTH(1-34) with high-frequency administration resulted in the formation of thin trabeculae, osteoclastogenesis, and accelerated bone remodeling, while low-frequency PTH administration showed a phenomenon of modeling-based formation through thicker trabeculae, mature osteoblasts, and new bone formation (Yamane *et al.*, 2017). This supports the notion that modeling-based bone gain can be an important axis in PTH anabolism and the primary principle of PTH as bone remodeling-driven bone anabolism.

The limitation of this study is that the therapeutic responses of rhPTH(1-34) and dimeric R^{25C} PTH(1-34) were focused on local surgical interventions, meaning that we could not investigate the central skeletal responses, such as in the femur and lumbar. Therefore, further research is needed to investigate these different responses by administering the PTH analogs at various frequencies and doses.

Overall, the study demonstrated the therapeutic effects of rhPTH(1-34) and dimeric R^{25C} PTH(1-34) on bone regeneration and titanium osseointegration using a beagle model with osteoporosis. Validation of the anabolic effects of rhPTH(1-34) and dimeric R^{25C} PTH(1-34) in large animals has

resulted in a broader understanding of their physiological and therapeutic functions and further expands their potential applications.

Material and Methods

Animal preparation

This study was conducted in compliance with the ARRIVE guidelines and approved by the Animal Research Committee of Cronex Co., Ltd., Hwaseong, South Korea. (CRONEX-IACUC 201801002). All animal experiments, including animal selection, management, preparation, and surgical protocols were conducted in compliance with the Ewha Womans University rules for animal experiments. The animals were housed in a standard laboratory environment ($21 \pm 1^\circ\text{C}$ with 40–70% humidity; 12Lhours light/dark cycle) with a standardized food/water supply.

Beagles were chosen for this study because the bone size and dentition could accommodate human dental implants and the application of mechanical force to implants. To induce osteoporosis, 12 female beagles underwent bilateral ovariectomy (OVX) at 12 weeks of age, followed by osteoporosis development for 12 weeks before being used for the next experiments at the age of 24 weeks. Health and oral hygiene were checked and maintained daily (**Figure 1A** [↗](#)).

Experiment protocol

Anesthesia was induced using Zolazepam/Tiletamine (10 mg/kg body weight, Zoletil; Virbac Laboratories, Carros, France) and xylazine hydrochloride (Rumpun[®], Bayer, Leverkusen, Germany), by intramuscular injection. The beagles were anesthetized using inhalation anesthesia for the implant surgery, and antibiotics (Ceftriaxone, Kyungdong Pharm, Seoul, South Korea) were administered for three days.

Both mandibular premolars 1–4 were extracted at 12 weeks after OVX. Then, 12 weeks later, three dental titanium implants (TS III[®] 3.0 x 10 mm, Osstem, Seoul, Korea) were inserted into the right lower jaw of each animal via a conventional implant surgical procedure under saline irrigation. Each implant was placed over at least a 3.0 mm distance. In detail, the first implant was placed over 3.0 mm behind the canine. The second and third implants were inserted to create a 3 mm circumferential bony defect using a Ø 6 mm trephine bur. Then, an additional bone graft (Bio-oss[®] small particle 1.0 g and Bio-gide[®], Geistlich, Switzerland) was placed on the bone defect around the 3rd implant, and the bone gap remained with nothing at the 2nd implant. On the left lower jaw, two artificial bone defects of 5 x 10 mm were made using trephine bur. The anterior hole remained defective without a graft, while a bone graft (Bio-oss[®] small particle 1.0 g and Bio-gide[®], Geistlich, Switzerland) was applied to the posterior hole.

After 2 weeks of healing, twelve female OVX beagles were randomly designated into three groups as follows: 1) control group with normal saline injection, 2) PTH(1-34) group with daily 40 µg/day rhPTH(1-34) injection (Forsteo[®], Eli Lilly and Company, Indianapolis, IN, USA), and 3) dimeric ^{R25C}PTH(1-34) group with 40 µg/day injection (dimeric ^{R25C}PTH(1-34), chemically synthesized by the by Anygen, Gwangju, Republic of Korea). Animals were injected subcutaneously for 10 weeks, after which, they were euthanized and the bone regeneration and implant osseointegration were evaluated (**Figure 1A** [↗](#), **B** [↗](#), and **C** [↗](#)). To analyze the dynamic bone formation, 10 mg/kg of calcein green (Sigma, St. Louis, USA) and 30 mg/kg of oxytetracycline yellow (Fluka, Shanghai, China) were intramuscularly administered at 2 weeks and 10 weeks after operation.

Micro-computed tomographic (μ CT) analysis

To examine the microarchitectural effects of PTH(1-34) and (Cys25)PTH(1-84) on bone regeneration and osseointegration, radiographic analysis was performed using μ CT on the right mandible (**Figure 1D** [↗](#)). The specimens were fixed in 4% paraformaldehyde for 48 hours before being assessed by micro-computed tomography (μ CT, SkyScan1173 ver. 1.6, Bruker-CT, Kontich, Belgium). The specimens were imaged with a pixel size of 29.83 μ m. The voltage and current intensities of the images were 130 kV and 60 μ A, respectively. The regions of interest (ROI) were determined as the 10 x 10 mm square area, located 3 mm from the bottom of the implant. Bone mineral density (BMD), bone volume (BV; mm^3), trabecular number (Tb.N; 1/mm), trabecular thickness (Tb.Th; μ m), and trabecular separation (Tb.S; μ m) were analyzed.

Histological and histomorphometric analysis

Bone histomorphometric parameters were computed and shown in accordance with recommendations by the ASBMR histomorphometric nomenclature committee. ([Dempster et al., 2013](#) [↗](#)) Goldner's trichrome and Masson trichrome staining were performed on both the right implantation and left bone defect sites, respectively. Specimens were dehydrated in increasing concentrations of ethanol and embedded in a mixture of ethanol and Technovit 7200 resin (Heraeus Kulzer, Wehrheim, Germany), with an increasing ratio of resin. Following resin infiltration, the specimens were hardened in a UV embedding system (KULZER EXAKT 520, Norderstedt, Germany) for a day. The undecalcified specimens were cut using an EXAKT diamond cutting system (EXAKT 300 CP, Norderstedt, Germany), and the soft tissue and bone were attached to an acryl slide by an adhesive system. The section width of the specimen was adjusted to $40 \pm 5 \mu\text{m}$ using a grinding system (EXAKT 400CS, KULZER, Norderstedt, Germany). The specimens on the right implantation site were stained with Goldner's trichrome and photographed by a Panoramic 250 Flash III system (3DHISTECH Ltd., Budapest, Hungary). The bone-implant contact ratio (BIC, %) was assessed as the linear percentage of the interface with direct contact between the bone and implant to the total interface of the implant using CaseViewer program software (3DHISTECH Ltd.).

Tartrate-resistant acid phosphatase (TRAP) assay was performed on the left bone defect sites. The bone specimens were fixed in 4% paraformaldehyde overnight and decalcified in 10% ethylenediaminetetraacetic acid (EDTA) for 7 days. The decalcifying solution was changed every other day. The specimens were embedded in paraffin and cut into sections. According to the manufacturer's instructions, the sections were deparaffinized and stained using a TRAP staining kit (Sigma, St. Louis, MO, USA). The number of TRAP-positive cells in the sections was counted under a microscope (DM2500, Leica Microsystems, Wetzlar, Germany).

Serum biochemical analysis

Fasting blood samples were drawn in the morning at baseline (T0; 12 weeks), the start of the injection (T1), and at euthanasia (T2) for P1NP (procollagen type I N-terminal propeptide), PTH (parathyroid hormone), CTx (C-terminal telopeptide), and calcium, and phosphorus. Calcium and phosphorus were analyzed using the Beckman AU480 Chemistry Analyzer (Beckman Coulter AU480). P1NP (procollagen type I N-terminal propeptide ELISA kit, Mybiosource), PTH (parathyroid hormone ELISA kit, Aviva Systems Biology), and CTX (C-terminal telopeptide (CTX-I) ELISA kit, Mybiosource) were analyzed by the ELISA method, according to the ELISA kit manufacturer's instructions. In all analyses, the measured values were below the limit of quantification for the standard curve.

Statistical analysis

Data for microarchitectural, histomorphometric, and serum biochemical analyses were expressed as mean and standard deviation (SD). Non-parametric tests were performed, including the Mann-Whitney and Kruskal-Wallis tests. Group differences in serum markers over time were compared

by repeatedly measuring the analysis of variance. Statistical analysis was performed using SPSS 26 (IBM Corp., USA) and Prism 10 (GraphPad, San Diego, CA, USA). P values of < 0.05 were set as statistically significant.

Acknowledgements

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Supplementary Figure. Three-dimensional reconstructed image of the bone surrounding the implants.

(A) rhPTH(1-34) (B) Dimeric R25CPTH(1-34) (C) Control

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Reviewer #1 (Public Review):

Summary:

This study, titled "Enhancing Bone Regeneration and Osseointegration using rhPTH(1-34) and Dimeric R25CPTH(1-34) in an Osteoporotic Beagle Model," provides valuable insights into the therapeutic effects of two parathyroid hormone (PTH) analogs on bone regeneration and osseointegration. The research is methodologically sound, employing a robust animal model and a comprehensive array of analytical techniques, including micro-CT, histological/histomorphometric analyses, and serum biochemical analysis.

Strengths:

The use of a large animal model, which closely mimics postmenopausal osteoporosis in humans, enhances the study's relevance to clinical applications. The study is well-structured, with clear objectives, detailed methods, and a logical flow from introduction to conclusion.

The findings are significant, demonstrating the potential of rhPTH(1-34) and dimeric R25CPH(1-34) in enhancing bone regeneration, particularly in the context of osteoporosis.

Weaknesses:

There are no major weaknesses.

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Reviewer #2 (Public Review):

Summary:

This article explores the regenerative effects of recombinant PTH analogues on osteogenesis.

Strengths:

Although PTH has known to induce the activity of osteoclasts, accelerating bone resorption, paradoxically its intermittent use has become a common treatment for osteoporosis. Previous studies successfully demonstrated this phenomenon in vivo, but most of them used rodent animal models, inevitably having a limitation. In this article, the authors tried to address this, using a beagle model, and assessed the osseointegrative effect of recombinant PTH analogues. As a result, the authors clearly observed the regenerative effects of PTH analogues, and compared the efficacy, using histologic, biochemical, and radiologic measurement for surgical-endocrinal combined large animal models. The data seem to be solid, and has potential clinical implications.

Weaknesses:

As PTH's mechanism has already been widely accepted, and the main focus of this article was to compare the preclinical efficacy of PTH analogues, the lack of detail biologic mechanism could be allowed. However, there are some suggestions to enhance the readability of the article:

First, the authors should clarify why they compared the effects of rhPTH(1-34) and of dimeric R25C2 PTH(1-34)? In most of the parameters, rhPTH(1-34) seems to be superior to dimeric R25C2 PTH(1-34). Why did the authors insist that the anabolic effects of dimer were prominent? Even though implication of dimeric R25C2 PTH(1-34) was drawn from genetic mutation studies, the authors should describe more clearly in the discussion the potential clinical benefits of the dimeric R25C2 PTH(1-34) compared to rhPTH(1-34), especially if dimeric R25C2 PTH(1-34) has just partial agonistic effect in pharmacodynamics.

Second, please describe the intermittent and continuous application of PTH analogues. Many of the readers may misunderstand that the authors' daily injection of PTHs were actually to mimic the clinical intermittent application or continuous one. Incorporation of the author's intention for experimental design would be more helpful for readers.

Third, please unify the nomenclature. Ensure consistency in the nomenclature throughout the article. Unify the naming conventions for PTH analogues, such as rhPTH(1-34) vs teriparatide and (Cys25)PTH(1-84) vs R25CPH(1-34) vs R25CPH(1-34) vs (1-84). Choose one nomenclature for each analogue and use it consistently throughout the article.

Overall, this paper is well-written, but these suggestions aim to improve clarity and consistency for a broader readership.

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Reviewer #3 (Public Review):**Summary:**

The work submitted by Dr. Jeong-Oh Shin and co-workers aims to investigate the therapeutic efficacy of rhPTH(1-34) and R25CPTH(1-34) on bone regeneration and osseointegration of titanium implants using a postmenopausal osteoporosis animal model.

In my opinion the findings presented are not strongly supported by the provided data since the methods utilized do not allow to significantly support the primary claims.

Strengths:

Strengths include certain good technologies utilized to perform histological sections (i.e. the EXAKT system).

Weaknesses:

Certain weaknesses significantly lower the enthusiasm for this work. Most important: the limited number of samples/group. In fact, as presented, the work has an n=4 for each treatment group. This limited number of samples/group significantly impairs the statistical power of the study. In addition, the implants were surgically inserted following a "conventional implant surgery", implying that no precise/guided insertion was utilized. This weakness is, in my opinion, particularly significant since the amount of bone osteointegration may greatly depend on the bucco-lingual positioning of each implant at the time of the surgical insertion (which should, therefore, be precisely standardized across all animals and for all surgical procedures).

On a minor note: not sure why the authors present a methodology to evaluate the dynamic bone formation (line 272) but do not present results (i.e. by means of histomorphometrical analyses) utilizing this methodology.

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